

Calculus and Linear Algebra

Vectors: Geometry, Coordinates and Operations

Lecture 2

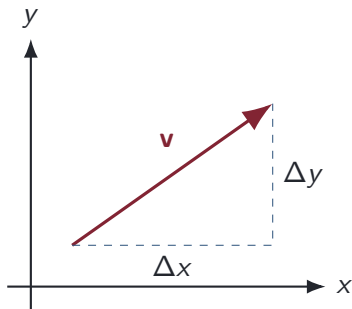
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18 September 2026

Why vectors?

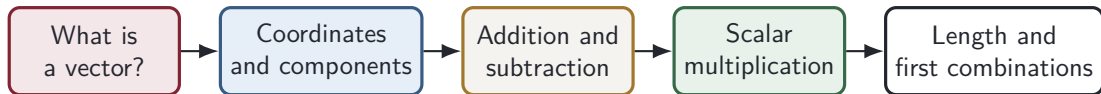
A single number can measure a quantity. A vector can describe several quantities **at once** – and, geometrically, a direction.

- ▶ movement in the plane;
- ▶ changes in several variables;
- ▶ bundles of goods;
- ▶ portfolios and allocations;
- ▶ later: systems of equations and multivariable calculus.



Today we learn to see the same vector as an arrow and as coordinates.

Today's route



The point

By the end of the lesson, an expression such as

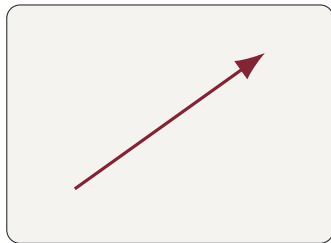
$$2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

should immediately suggest both an **algebraic calculation** and a **geometric picture**.

Warm-up: which of these needs direction?

Which quantities can be described completely by just one number?

1. Temperature: 24°C
2. Mass: 3 kg
3. A movement of 5 m east
4. A force pulling up and right



Scalar versus vector

Scalar

A scalar has **magnitude only**.

5, -2.4, 100

Examples: price, temperature, time, mass.

scalar = one number

Vector

A vector has several components and, geometrically, a **magnitude and direction**.

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Examples: displacement, velocity, a bundle (x_1, x_2) .

vector = ordered components

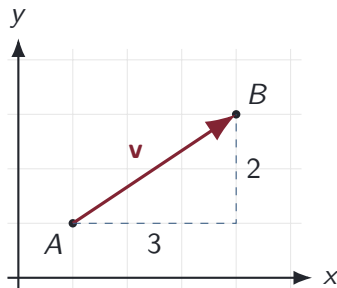
A vector as a displacement

Start at $A = (1, 1)$ and finish at $B = (4, 3)$.
The displacement is

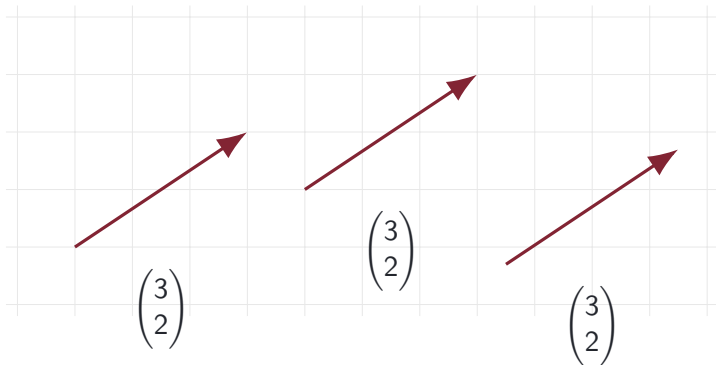
$$\mathbf{v} = B - A = \begin{pmatrix} 4 - 1 \\ 3 - 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

It says:

3 units to the right, 2 units up.



The same vector can be drawn in different places



The three arrows have the

same **length**, the same **direction**, and the same **sense**. Therefore they represent the same vector. A vector is not tied to one particular starting point.

Coordinates: the algebraic description

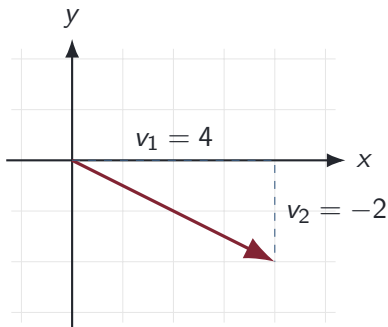
In $\mathbb{R}^2$ we write

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

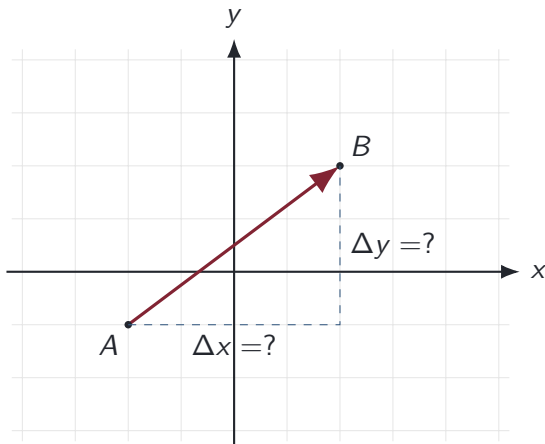
The numbers v_1 and v_2 are the **components** of the vector. For

$$\mathbf{v} = \begin{pmatrix} 4 \\ -2 \end{pmatrix},$$

we move 4 units right and 2 units down.



Read a vector from a graph



Find the vector from A to B without looking at the next slide.

Read a vector from a graph: check

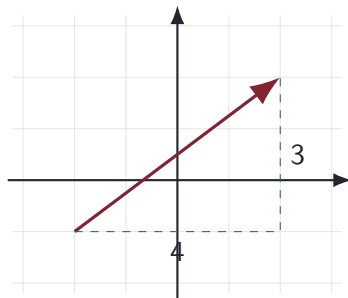
From $A = (-2, -1)$ to $B = (2, 2)$:

$$\Delta x = 2 - (-2) = 4,$$

$$\Delta y = 2 - (-1) = 3.$$

Therefore

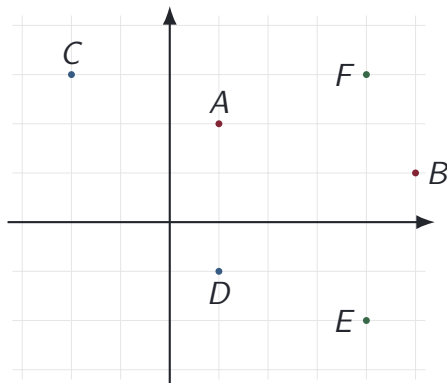
$$\boxed{\vec{AB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}}.$$



Your turn: components

For each pair of points, compute the vector from the first point to the second.

1. $A = (1, 2)$, $B = (5, 1)$
2. $C = (-2, 3)$, $D = (1, -1)$
3. $E = (4, -2)$, $F = (4, 3)$



Equality of vectors

Two vectors are equal when their corresponding components are equal:

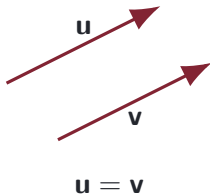
$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} c \\ d \end{pmatrix} \iff \begin{cases} a = c, \\ b = d. \end{cases}$$

Example:

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} x+1 \\ 2y+1 \end{pmatrix}$$

means

$$x+1=2, \quad 2y+1=-1.$$



Two special vectors

Zero vector

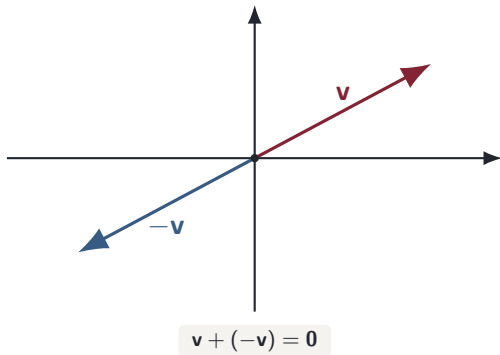
$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

It represents no displacement.

Opposite vector

$$\mathbf{v} = \begin{pmatrix} a \\ b \end{pmatrix} \implies -\mathbf{v} = \begin{pmatrix} -a \\ -b \end{pmatrix}.$$

It has the same length and the opposite direction.



Quick checkpoint: vector notation

Answer mentally first.

1. If $\mathbf{v} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$, what is $-\mathbf{v}$?

2. Is $\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$?

3. Find x, y if

$$\begin{pmatrix} x - 1 \\ y + 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}.$$

Components have positions: first component with first component, second with second.

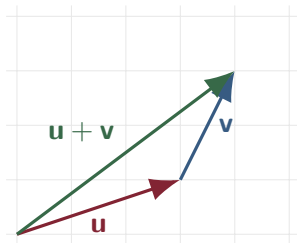
Adding vectors: the head-to-tail rule

Suppose

$$\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

To add them geometrically:

1. draw $\mathbf{u}$;
2. place the tail of $\mathbf{v}$ at the head of $\mathbf{u}$;
3. connect the starting point to the final point.



Addition in coordinates

Vector addition is component by component:

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a + c \\ b + d \end{pmatrix}.$$

For

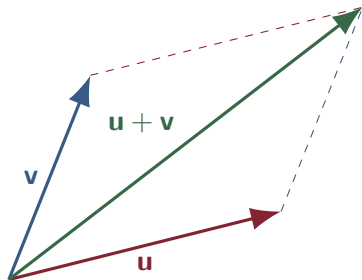
$$\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

we get

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 + 1 \\ 1 + 2 \end{pmatrix} = \boxed{\begin{pmatrix} 4 \\ 3 \end{pmatrix}}.$$

The coordinate calculation and the head-to-tail picture must agree.

The parallelogram rule



When two vectors start at the same point, their sum is

the **diagonal of the parallelogram**.

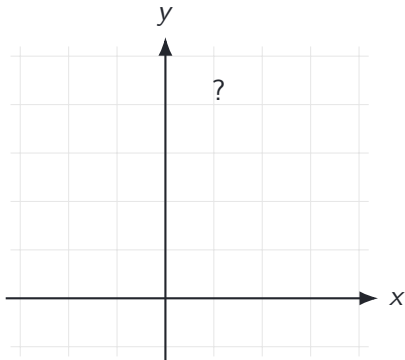
Same operation, different geometric construction.

Exercise: add the vectors

Let

$$\mathbf{u} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

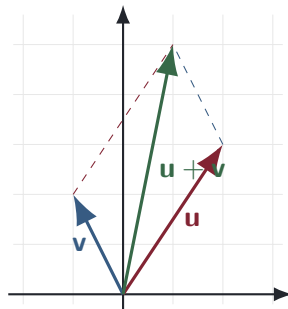
1. Compute $\mathbf{u} + \mathbf{v}$.
2. Draw $\mathbf{u}$ and $\mathbf{v}$ from the origin.
3. Draw the sum using the parallelogram rule.



Exercise: graphical check

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \boxed{\begin{pmatrix} 1 \\ 5 \end{pmatrix}}.$$

Notice that the negative first component of $\mathbf{v}$ pulls the sum to the left.



Subtraction: add the opposite

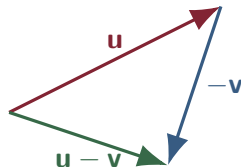
$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}).$$

If

$$\mathbf{u} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 3 \end{pmatrix},$$

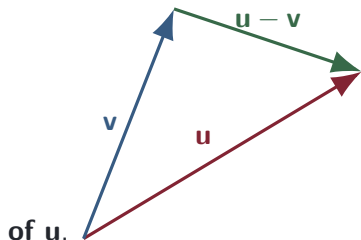
then

$$\mathbf{u} - \mathbf{v} = \begin{pmatrix} 4 - 1 \\ 2 - 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$



Another geometric meaning of $\mathbf{u} - \mathbf{v}$

If $\mathbf{u}$ and $\mathbf{v}$ start at the same point, then $\mathbf{u} - \mathbf{v}$ points **from the head of $\mathbf{v}$ to the head**



This picture will be useful whenever we compare two positions or two data points.

Exercise: addition and subtraction

Let

$$\mathbf{a} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbf{a} + \mathbf{b}$

2. $\mathbf{a} - \mathbf{b}$

3. $\mathbf{b} - \mathbf{a}$

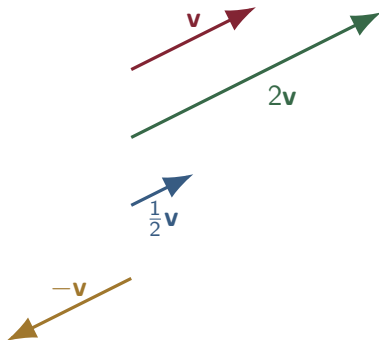
4. $-\mathbf{a} + \mathbf{b}$

Your turn Which two answers are opposites of each other? Why must that happen?

Scalar multiplication: stretch, shrink, reverse

For a scalar λ ,

$$\lambda \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \lambda a \\ \lambda b \end{pmatrix}.$$

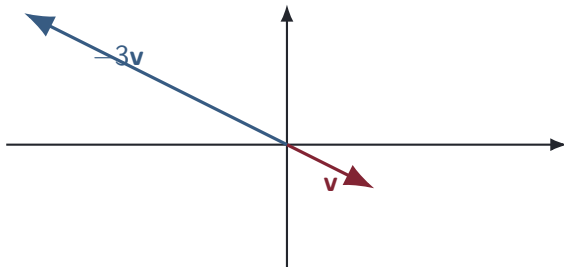


What the sign of the scalar does

- ▶ $\lambda > 1$: same direction, longer.
- ▶ $0 < \lambda < 1$: same direction, shorter.
- ▶ $\lambda < 0$: opposite direction.
- ▶ $\lambda = 0$: the zero vector.

Example:

$$-3 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \end{pmatrix}.$$



Your turn: scalar multiplication

Let

$$\mathbf{v} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Compute and sketch:

1. $2\mathbf{v}$

2. $\frac{1}{2}\mathbf{v}$

3. $-\mathbf{v}$

4. $-\frac{3}{2}\mathbf{v}$

Before calculating, predict the direction and relative length.

Combine the operations

Vector algebra follows familiar rules.

$$2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} - 3 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} - \begin{pmatrix} -3 \\ 3 \end{pmatrix} = \boxed{\begin{pmatrix} 5 \\ -7 \end{pmatrix}}.$$

A useful habit

Do not rush. Separate the operations:

scalar multiplication $\longrightarrow$ addition/subtraction $\longrightarrow$ simplify.

Exercise: a little vector algebra

Given

$$\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix},$$

compute:

1. $3\mathbf{u}$

2. $2\mathbf{v} - \mathbf{u}$

3. $2\mathbf{u} + 3\mathbf{v}$

4. $-2(\mathbf{u} - \mathbf{v})$

Your turn For item 3, estimate the direction of the answer before computing it.

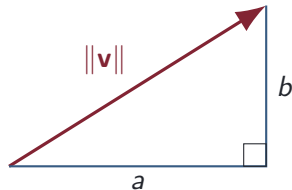
Length of a vector

For

$$\mathbf{v} = \begin{pmatrix} a \\ b \end{pmatrix},$$

the length (or Euclidean norm) is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2}.$$



This is just the Pythagorean theorem.

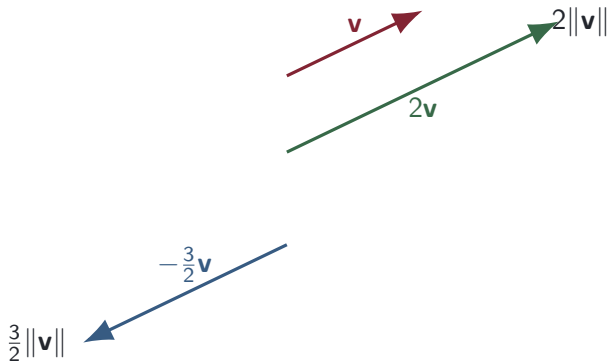
Example:

$$\left\| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = \sqrt{3^2 + 4^2} = 5.$$

Length: what changes under scaling?

Suppose $\mathbf{v}$ has length $\|\mathbf{v}\|$. Then

$$\|\lambda \mathbf{v}\| = |\lambda| \|\mathbf{v}\|.$$



Exercise: lengths

Compute the length of each vector.

1. $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$

2. $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$

3. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

4. $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$

Then answer without recalculating from scratch:

$$\text{if } \|\mathbf{v}\| = 5, \quad \|-3\mathbf{v}\| = ?$$

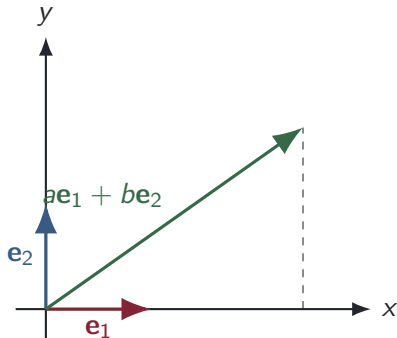
Unit vectors: the coordinate directions

The standard unit vectors in $\mathbb{R}^2$ are

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

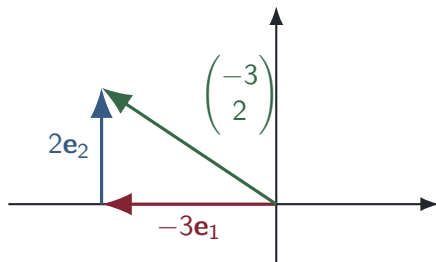
They have length 1 and point along the coordinate axes. Every vector can be written as

$$\begin{pmatrix} a \\ b \end{pmatrix} = a\mathbf{e}_1 + b\mathbf{e}_2.$$



Decompose a vector into coordinate directions

$$\begin{pmatrix} -3 \\ 2 \end{pmatrix} = -3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -3\mathbf{e}_1 + 2\mathbf{e}_2.$$



Coordinates are exactly the weights of the two basic directions.

Exercise: from components to geometry

For each vector:

1. write it using $\mathbf{e}_1$ and $\mathbf{e}_2$;
2. sketch it from the origin;
3. state the quadrant in which its head lies.

$$\begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

Your turn Which one has the greatest length?

Application: total displacement

A person walks 4 km east and 1 km north, then 2 km west and 3 km north.

First displacement:

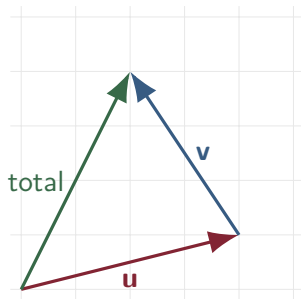
$$\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

Second displacement:

$$\mathbf{v} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Total:

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$



Application: a bundle of goods is a vector

A consumer bundle with two goods can be written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

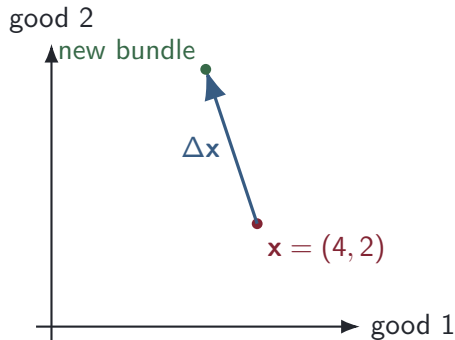
For example,

$$\mathbf{x} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

means 4 units of good 1 and 2 units of good 2. A change

$$\Delta \mathbf{x} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Good 1 decreases by 1; good 2 increases by 3.



Exercise: update the bundle

A consumer starts from

$$\mathbf{x} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}.$$

The change in quantities is

$$\Delta \mathbf{x} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

1. What is the new bundle $\mathbf{x} + \Delta \mathbf{x}$?
2. Which good increased? Which decreased?
3. Draw the initial and final bundles as points in the plane.

A first look at linear combinations

We already know how to calculate expressions of the form

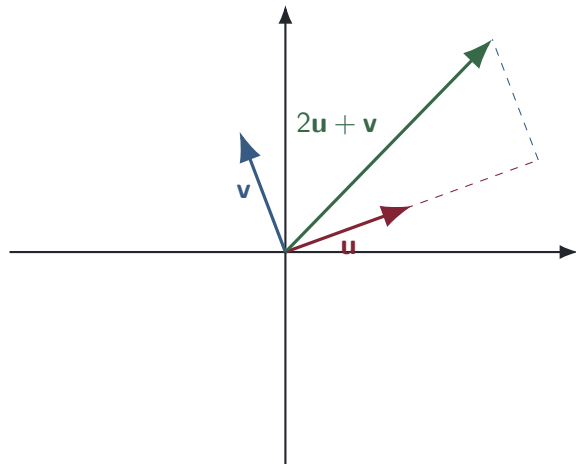
$$\lambda \mathbf{u} + \mu \mathbf{v}.$$

Such an expression is called a **linear combination** of $\mathbf{u}$ and $\mathbf{v}$. Example:

$$2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Next lecture: what points can we reach by varying the coefficients?

Linear combinations are movements along directions



The coefficients tell us how much of

each direction we use.

Exercise: find the coefficients

Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Find λ and μ such that

$$\lambda \mathbf{u} + \mu \mathbf{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Then repeat for

$$\begin{pmatrix} -4 \\ 5 \end{pmatrix}.$$

Your turn What do λ and μ represent here?

Mixed practice: choose the operation

Let

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}.$$

1. $\mathbf{u} + \mathbf{v}$

2. $\mathbf{u} - \mathbf{v}$

3. $3\mathbf{u}$

4. $2\mathbf{u} - \mathbf{v}$

5. $\|\mathbf{u}\|$

6. $\| -2\mathbf{v} \|$

7. vector from $A = (1, 3)$ to $B = (-2, 5)$

8. x, y if $\begin{pmatrix} x+1 \\ 2y \end{pmatrix} = \mathbf{v}$

7 minutes – draw a small sketch whenever it helps.

Mixed practice: quick check

1. $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$

2. $\begin{pmatrix} 5 \\ -5 \end{pmatrix}$

3. $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$

4. $\begin{pmatrix} 7 \\ -6 \end{pmatrix}$

5. $\sqrt{5}$

6. $2\sqrt{25} = 10$

7. $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

8. $x = -4, y = 2$

Check If an answer is wrong, locate the operation where the sign changed.

Challenge: the missing corner of a parallelogram

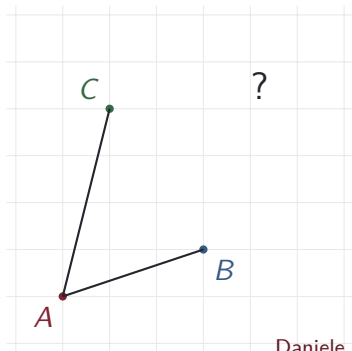
Three vertices of a parallelogram are

$$A = (1, 1), \quad B = (4, 2), \quad C = (2, 5).$$

Assume A is the common vertex of the two sides AB and AC .

Find the fourth vertex D . Hint:

$$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC}.$$



Challenge: solution

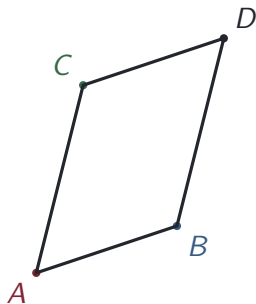
$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Therefore

$$\overrightarrow{AD} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

Since $A = (1, 1)$,

$$D = A + \overrightarrow{AD} = (1, 1) + (4, 5) = \boxed{(5, 6)}.$$



Common mistakes to avoid

Mistake 1: adding across components

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \end{pmatrix} \neq \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \quad \text{but} \quad \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Mistake 2: forgetting both signs

$$-\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Mistake 3: confusing point and vector

$$B - A = \overrightarrow{AB}.$$

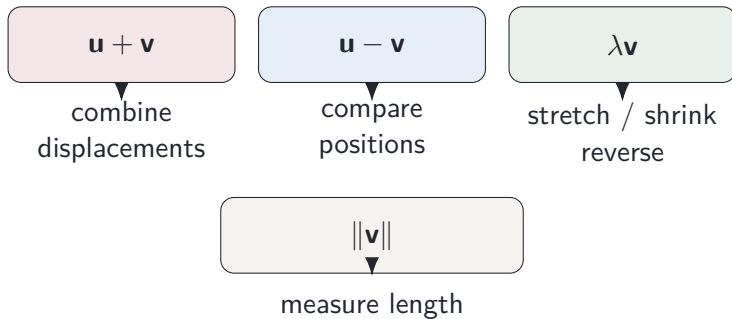
The coordinates look similar, but the meanings differ.

Mistake 4: length is not a component sum

$$\left\| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = 5 \neq 7.$$

When in doubt, draw the arrow. Geometry catches many sign errors.

A visual summary of the operations



Five ideas to leave with today

1. A vector in $\mathbb{R}^2$ is described by two components.
2. Geometrically, a vector represents a displacement with length and direction.
3. Addition and subtraction are performed component by component and have clear geometric meanings.
4. Multiplying by a scalar changes length and possibly direction.
5. Expressions such as $\lambda \mathbf{u} + \mu \mathbf{v}$ are linear combinations – the starting point of the next lecture.

Always translate: coordinates $\leftrightarrow$ arrows.

Exit ticket

Before you leave, answer these without notes.

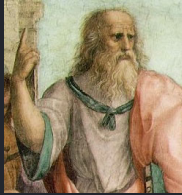
1. Find the vector from $A = (-1, 2)$ to $B = (3, 0)$.

2. Compute

$$2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} - \begin{pmatrix} -2 \\ 4 \end{pmatrix}.$$

3. What happens geometrically when a vector is multiplied by $-\frac{1}{2}$?

Next lecture: **linear combinations, vector spaces and subspaces.**



“Mathematics is the language
in which the gods speak to
people.”

Plato

